Linux Kernel Programming

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Agenda

- I2C Driver

I2C Driver – Handson using SSD1306

- SSD1306 is a single-chip CMOS OLED/PLED driver with a controller for an organic / polymer light-emitting diode dot-matrix graphic display system. It consists of 128 segments and 64commons.



I2C Driver – Handson using SSD1306

- The SSD1306 embeds with contrast control, display RAM, and oscillator, which reduces the number of external components and power consumption. It has 256-step brightness control. Data/Commands are sent from general MCU through the hardware selectable 6800/8000 series compatible Parallel Interface, I2C interface, or Serial Peripheral Interface. It is suitable for many compact portable applications, such as mobile phone sub-display, MP3 player and calculator, etc.

- The operating voltage of the SSD1306 controller is from 1.65V to 3.3V while the OLED panel requires 7V to 15V supply voltage. All these different power requirements are sufficed using internal charge pump circuitry. This makes it possible to connect it to any 5V logic microcontroller easily without using any logic level converter.

- Regardless of the size of the OLED module, the SSD1306 driver has a built-in 1KB Graphic Display Data RAM (GDDRAM) for the screen which holds the bit pattern to be displayed. This 1K memory area is organized in 8 pages (from 0 to 7). Each page contains 128 columns/segments (block 0 to 127). And each column can store 8 bits of data (from 0 to 7). That surely tells us we have

8 pages x 128 segments x 8 bits of data = 8192 bits = 1024 bytes = 1KB memory

I2C Driver – Handson using SSD1306

- Here are the complete specifications:

Display Technology	OLED (Organic LED)
MCU Interface	I2C / SPI
Screen Size	0.96 Inch Across
Resolution	128×64 pixels
Operating Voltage	3.3V – 5V
Operating Current	20mA max
Viewing Angle	160°
Characters Per Row	21
Number of Character Rows	7

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- There are two types of data that we can send to the SSD1306 OLED.
  - Command
  - Data to be written into the GDDRAM

Whenever you send some data, you have to send the control byte first, then send the data byte after that. That control byte used to tell the data that you are sending is command or data.

Control Byte

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Co bit *	D/C *	0	0	0	0	0	0

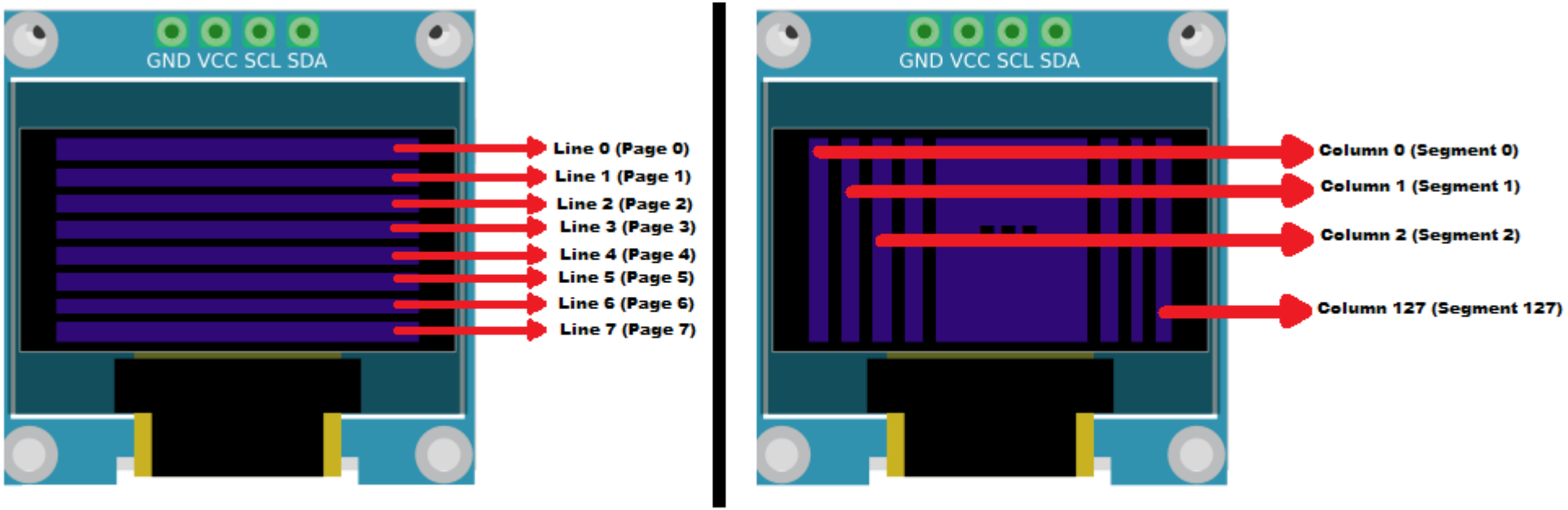
The first 6 bits should be 0.

D/C – If this bit is 1, then the next byte will be a command. If this bit is 0, then the next byte will be data.

Co – If this bit is 0, then the following bytes contain data bytes only.

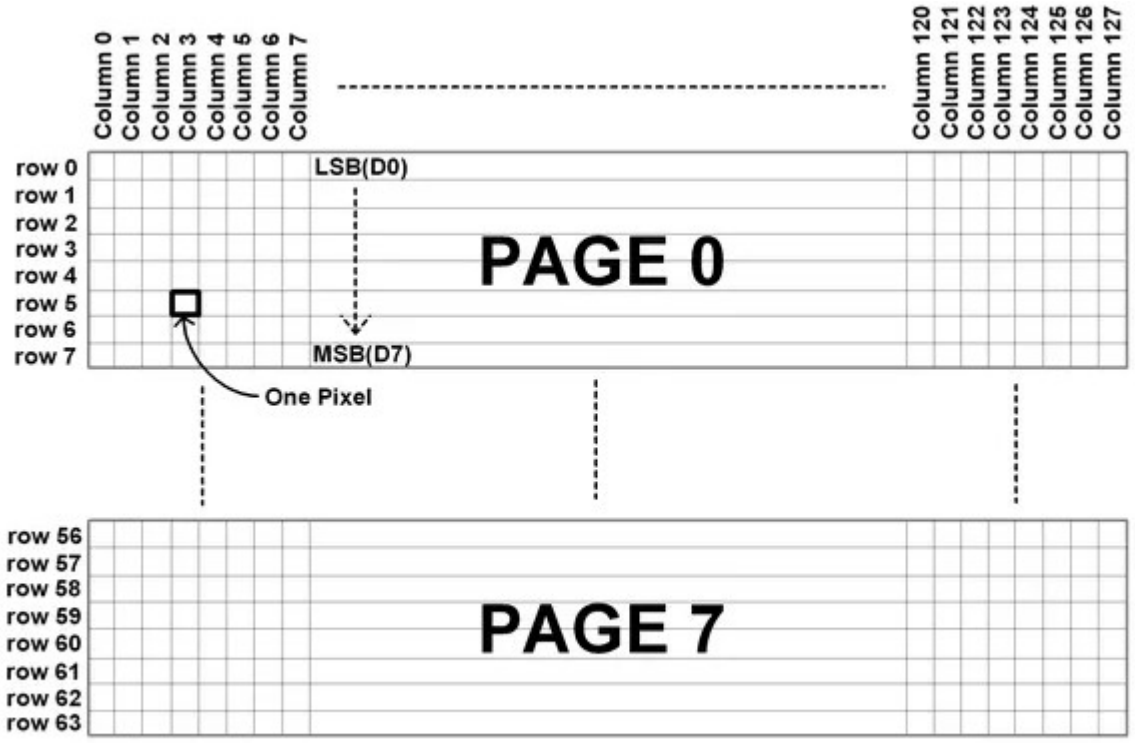
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- Basically SSD1306 has 8 lines (pages) and 128 columns(segments). Please refer to the below image for a clear understanding.



I2C Driver – Handson using SSD1306

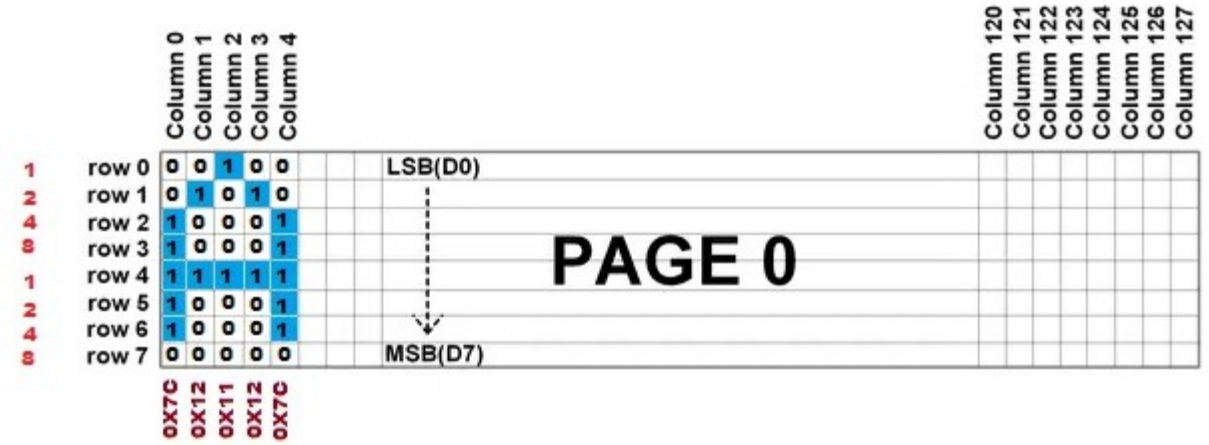
- Each page contains 8 rows. So, a total of 64 rows will be present in the SSD1306's GDDRAM (Graphics Display Data RAM). The GDDRAM (Graphics Display Data RAM) is a bitmapped static RAM. It holds the bit pattern to be displayed. For understanding the rows, please look at the image below.



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- As per our understanding, it has 128 columns and 64 rows. So the size of GDDRAM is 128 x 64 bits and it is divided into eight pages, from PAGE0 to PAGE7, which are used for monochrome 128×64 dot matrix display

- we are going to write one character in 5×8. The below image explains how to write our characters.



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- There are many commands available. Some commands are given below.

Command	Command	Description
0x00	COMMAND	This command is used to indicate the next data byte is acted as a command.
0x40~0x7F	Display Start Line	This command sets the Display Start Line register to determine the starting address of display RAM
0xAE	Display OFF	This command is used to turn OFF the OLED display panel.
0xAF	Display ON	This command is used to turn ON the OLED display panel.
0x20	Memory Addressing Mode	If horizontal address increment mode is enabled by command 20h, after finishing read/write one column data, it is incremented automatically to the next column address.
0x26/0x27	Continuous Horizontal Scroll	0x26 – Right horizontal scroll 0x27 – Left horizontal scroll
0x29/0x2A	Continuous Vertical and Horizontal Scroll	0x29 – Vertical and Right Horizontal Scroll 0x2A – Vertical and Left Horizontal Scroll
0x2E	Deactivate Scroll	This command deactivates the scroll.
0x2F	Activate Scroll	This command activates the scroll if it is configured before.

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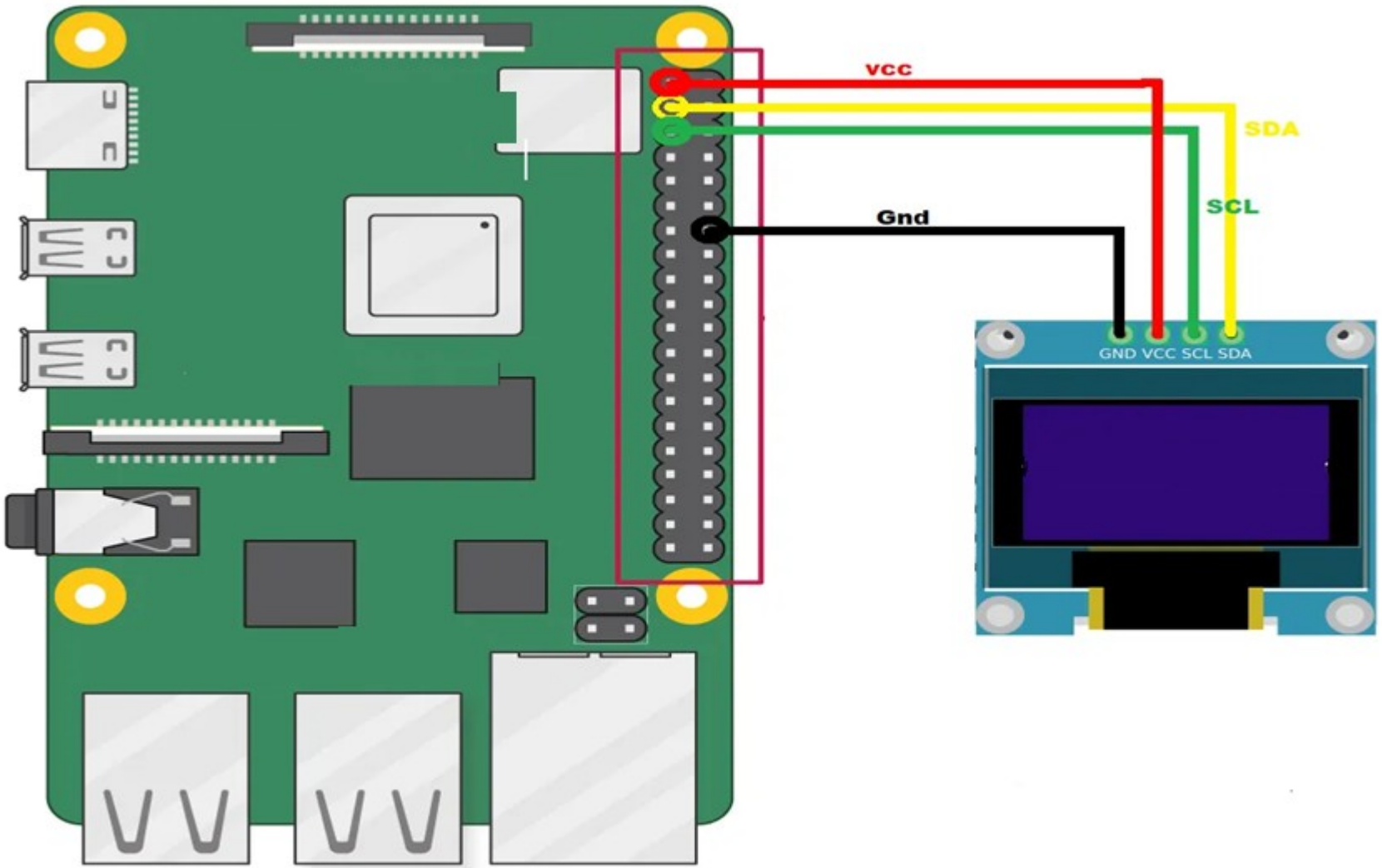
Command	Command	Description
0x21	Column Address	This command is used to define the current read/write column address in graphic display data RAM.
0x22	Page Address	This command is used to define the current read/write Line(Page as per data sheet) address in graphic display data RAM.
0x81	Contrast Control	This command sets the Contrast Setting of the display. The chip has 256 contrast steps from 00h to FFh. The segment output current increases as the contrast step value increases.
0xA0	Segment Re-map	This command sets the column address 0 is mapped to SEG0.
0xA1	Segment Re-map	This command sets the column address 127 is mapped to SEG0.
0xA6	Normal Display	This command sets the display to normal mode. In the normal display, a RAM data of 1 indicates an “ON” pixel.
0xA7	Inverse Display	This command sets the display to inverse mode. In the normal display, a RAM data of 0 indicates an “ON” pixel.
0xA8	Multiplex Ratio	This command sets the display multiplex ratio.

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Command	Command	Description
0xC0/0xC8	COM Output Scan direction	0xC0 – Normal mode (RESET) Scan from COM0 to COM[N –1] 0xC8- Remapped mode. Scan from COM[N-1] to COM0 Where N is the Multiplex ratio.
0xD3	Display Offset	This command sets vertical shift by COM from 0d~63d.
0xD5	Display Clock Divide Ratio/Oscillator Frequency	This command sets the display Clock Divide Ratio and Oscillator Frequency.

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- In this example, we are going to use the SSD1306 OLED display as the slave device.

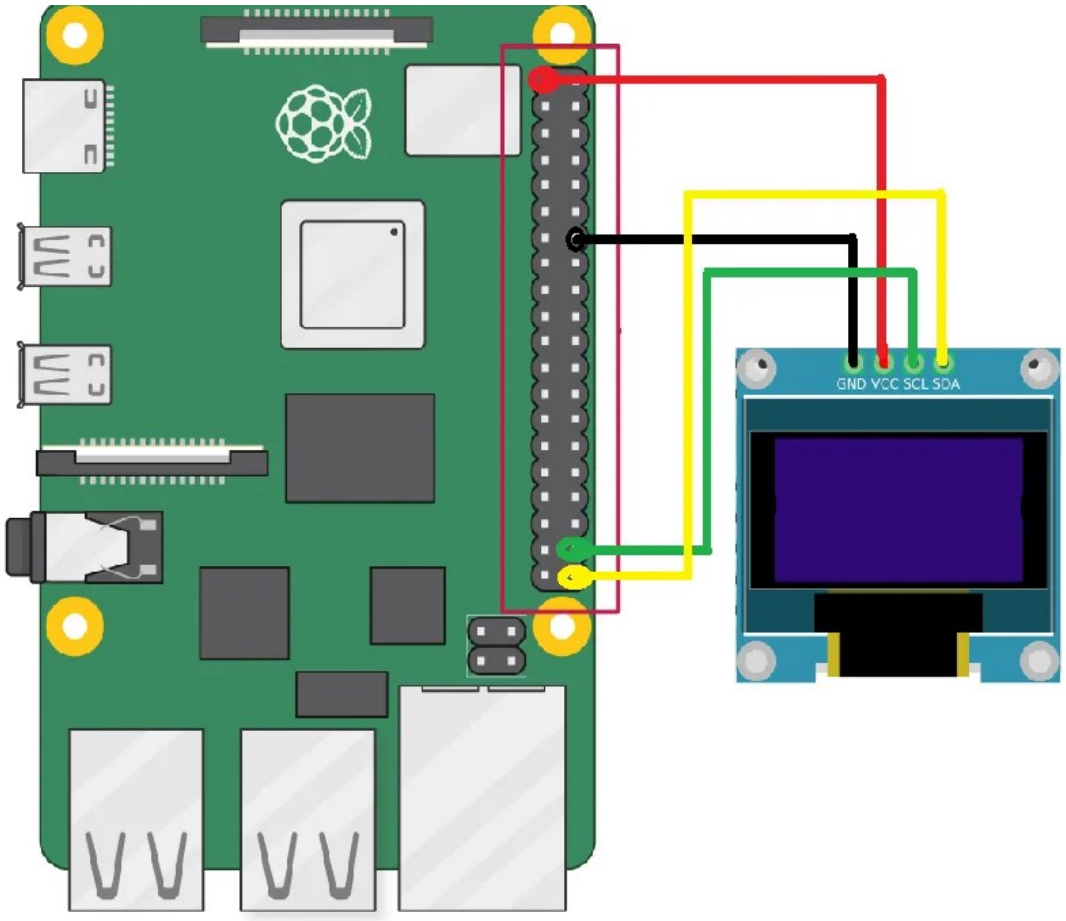


I2C Driver (Bit Banging Method) – GPIO (Method2)

Connection Diagram

SCL – GPIO 20

SDA – GPIO 21



Thank you